PROJECT PROMPT: RWANDA ROAD TRAFFIC RULES AND SAFETY EDUCATION SYSTEM

Act as a senior software engineer, UI/UX designer, database architect, and full-stack developer.

Build a modern, professional, responsive web application called **Amategeko y'Umuhanda (Rwanda Road Traffic Rules and Safety Education System)**.

1. Project Purpose

Develop a digital platform that helps Rwandans learn road traffic rules, understand road signs, prepare for driving theory examinations, and improve road safety awareness.

2. Main Features

- **User Registration and Login:** Admin, Instructor, and Learner accounts.
- **Road Signs:** Display traffic signs with images, names, meanings, categories, and explanations.
- **Traffic Rules:** Teach road regulations, speed limits, road priorities, overtaking, parking, and pedestrian safety.
- **Learning Materials:** Text lessons, images, videos, and practical examples.
- **Online Examinations:** Multiple-choice questions, timed tests, automatic marking, scores, and explanations.
- **Practice Tests:** Allow learners to practise unlimited questions.
- **Progress Tracking:** Display completed lessons, examination history, and performance.
- **Road Safety Education:** Provide accident prevention guidance and emergency procedures.
- **Admin Dashboard:** Manage users, lessons, questions, road signs, and reports.
- **Notifications:** Inform learners about new lessons and examination results.

3. Technology Stack

- Backend: Python and Django.
- Frontend: HTML5, CSS3, JavaScript, Bootstrap 5.
- Database: PostgreSQL.

- Charts: Chart.js.
- Icons: Font Awesome.
- Version Control: Git and GitHub.

4. Design Requirements

Create a modern, attractive, professional interface inspired by leading educational platforms.

Use:

- Red, white, dark blue, and light grey.
- Professional navigation bar.
- Attractive homepage with a hero section.
- Interactive road sign cards.
- Clean dashboard with statistics.
- Responsive design for mobile, tablet, and desktop.
- Simple and user-friendly navigation.

5. Main Pages

1. Homepage
2. About Us
3. Road Signs
4. Traffic Rules
5. Learning Materials
6. Practice Examination
7. Online Examination
8. Learner Dashboard
9. Login and Registration
10. Admin Dashboard
11. Contact Us

6. Legal Accuracy

Use the current applicable road traffic legislation of Rwanda and official Rwandan road safety guidance. Verify every legal rule, penalty, speed limit, and examination requirement against authoritative sources. Do not invent regulations or present outdated legislation as current.

7. Development Instructions

Build the application step by step:

1. Set up the Django project.
2. Design the database models.
3. Implement authentication and user roles.
4. Develop the homepage and navigation.
5. Develop road signs and traffic rules modules.
6. Develop the learning and examination modules.
7. Develop dashboards and reports.
8. Test all functionalities.
9. Improve security and responsiveness.
10. Prepare the application for deployment.

Write clean, reusable, well-documented code. Do not provide incomplete mockup pages. All buttons, forms, menus, and examination features must work correctly.

The final result should be a complete, functional, professional Rwanda Road Traffic Rules and Safety Education System.